

Task 1 – Reconnaissance and Vulnerability Assessment

Objective

The aim of this task is to conduct active reconnaissance and vulnerability analysis on a controlled test system using industry-standard scanning tools. The goal is to identify exposed services, open ports, and potential security issues.

Tools Utilized

- Nmap
- Nikto
- Kali Linux

Target Details

- Host: scanme.nmap.org
- IP Address: 45.33.32.156

Overview

Reconnaissance and vulnerability scanning are essential steps in cybersecurity testing. These processes help in discovering system exposure points such as open ports, running services, and configuration weaknesses that could be exploited by attackers.

In this assessment, Nmap was used for network discovery and service enumeration, while Nikto was employed to analyze web server vulnerabilities.

Phase 1: Network Scanning with Nmap

Command Executed

```
nmap -sV -T4 scanme.nmap.org
```

Purpose

- Detect open ports
- Identify active services
- Determine service versions

Findings

- 22/tcp – SSH (OpenSSH)
- 80/tcp – HTTP (Apache Web Server)
- 9929/tcp – Nping Echo Service
- 31337/tcp – Unknown (tcpwrapped)

Result

```
# Nmap 7.98 scan initiated Thu Apr 2 08:39:47 2026 as:
/usr/lib/nmap/nmap --privileged -sV -T4 -oN /home/kali/Desktop/nmap.txt scanme.nmap.org

Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.12s latency).

PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
179/tcp    filtered  bgp
445/tcp    filtered  microsoft-ds
646/tcp    filtered  ldp
9929/tcp   open      nping-echo    Nping echo
31337/tcp  open      tcpwrapped

Service Info: OS: Linux

# Nmap done: 1 IP address (1 host up) scanned in 98.35 seconds
```

Analysis

The presence of multiple open ports indicates potential entry points into the system. Each exposed service increases the attack surface and requires proper monitoring and configuration.

Phase 2: Web Vulnerability Scanning with Nikto

Command Executed

```
nikto -h scanme.nmap.org
```

Purpose

- Identify outdated software
- Detect misconfigurations
- Check missing security headers
- Discover known vulnerabilities

Findings

- Apache server information disclosure
- Missing HTTP security headers
- Apache mod_negotiation enabled
- Possible outdated Apache version

Result

```
- Nikto v2.6.0
-----
+ Target Host: scanme.nmap.org
+ Target Port: 80

+ GET /index: Uncommon header 'tcn' found.
+ Apache mod_negotiation is enabled with MultiViews.
+ Attackers may brute force file names.
+ Alternative file found: index.html

+ Apache/2.4.7 appears to be outdated
+ Current secure version is at least 2.4.66

+ Nikto scan completed
```

Analysis

These findings indicate weaknesses in the web server configuration. Missing security headers can expose the system to attacks such as cross-site scripting (XSS) and clickjacking.

Identified Security Risks

- Open SSH Port – Medium risk (brute-force attacks)
 - Outdated Apache Version – High risk (known exploits)
 - Missing Security Headers – Medium risk (web attacks)
 - Exposed Services – Medium risk (information disclosure)
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Recommendations

- Update Apache server to latest version
 - Implement HTTP security headers
 - Restrict unnecessary open ports
 - Configure firewall rules
 - Perform regular vulnerability scans
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Conclusion

This task demonstrated how reconnaissance and vulnerability scanning tools can be used to identify open ports, services, and vulnerabilities. Proper security measures and continuous monitoring are essential to reduce risks.

Ethical Consideration

All activities were conducted on authorized systems strictly for educational purposes only.